

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$$

$$(n+1)! = \underbrace{1 \cdot 2 \cdot 3 \cdot \dots \cdot n}_{n!} \cdot (n+1) = n! \cdot (n+1)$$

$$(n+3)! = (n+2)! \cdot (n+3) = n! \cdot (n+1) \cdot (n+2) \cdot (n+3)$$

$$0! = 1$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

SYMBOL DWUMIANOWY NEWTONA

$\binom{n}{k}$ - czytamy „n po k” lub „n nad k”

$$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$$

Np. $\binom{7}{3} = \frac{7!}{(7-3)! \cdot 3!} = \frac{7!}{4! \cdot 3!} = \frac{\cancel{4!} \cdot \cancel{5} \cdot \cancel{6} \cdot 7}{\cancel{4!} \cdot 1 \cdot 2 \cdot 3} = 35$

$$\binom{1}{1} = \frac{1!}{(1-1)! \cdot 1!} = \frac{1}{0! \cdot 1} = \frac{1}{1 \cdot 1} = 1$$

$$\binom{n}{n} = \frac{\cancel{n!}^1}{(n-n)! \cdot \cancel{n!}_1} = \frac{1}{1} = 1$$

$$\binom{n}{0} = \frac{n!}{(n-0)! \cdot 0!} = \frac{\cancel{n!}^1}{\cancel{n!} \cdot 1} = 1$$

$$\binom{n}{1} = \frac{n!}{(n-1)! \cdot 1!} = \frac{\cancel{(n-1)!}^n \cdot n}{\cancel{(n-1)!}} = n$$

TRÓJKĄT PASCALA

$\binom{n}{k}$ — "wiersze"
 $\binom{n}{k}$ — "kolumny"

$$\begin{array}{c} \binom{0}{0} \\ \binom{1}{0} \quad \binom{1}{1} \\ \binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \end{array}$$

$$\begin{array}{ccccccc} & & & & & & n=0 \\ & & & & & & 1 \\ & & & & & 1 & 1 & n=1 \\ & & & 1 & 2 & 1 & & n=2 \\ & & 1 & 3 & 3 & 1 & & n=3 \\ & 1 & 4 & 6 & 4 & 1 & & n=4 \end{array}$$

$k=0 \quad k=1 \quad k=2 \quad k=3 \quad k=4$

$$\binom{4}{2} = 6$$

$$\binom{5}{3} = 10$$

$$\binom{3}{2} = 3$$

$$\begin{array}{ccccc} 1 & 4 & 6 & 4 & 1 \\ 1 & 5 & 10 & 10 & 5 & 1 \end{array}$$

$k=0 \quad k=1 \quad k=2 \quad k=3 \quad k=4$

Własność:

$$\binom{n}{k} = \binom{n}{n-k}$$

WZÓR DWUMIANOWY NEWTONA

$$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} a^0 b^n$$

$\begin{smallmatrix} n \\ 1 \end{smallmatrix}$ $\begin{smallmatrix} n \\ n \end{smallmatrix}$ $\begin{smallmatrix} n \\ 1 \end{smallmatrix}$

Np. $(a+b)^3 = \binom{3}{0} a^3 b^0 + \binom{3}{1} a^2 b^1 + \binom{3}{2} a^1 b^2 + \binom{3}{3} a^0 b^3 =$

$$= a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = [a + (-b)]^3 = a^3 + 3a^2(-b) + 3a(-b)^2 + (-b)^3 =$$
$$= a^3 - 3a^2b + 3ab^2 - b^3$$

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$(\square + \Delta)^2 = \square^2 + 2 \cdot \square \cdot \Delta + \Delta^2$$

$$(3x + 5y)^2 = (3x)^2 + 2 \cdot 3x \cdot 5y + (5y)^2 = 9x^2 + 30xy + 25y^2$$

$$(4a + 7b)^2 = 16a^2 + 56ab + 49b^2$$

$$(a \pm b)^2 = a^2 \pm 2ab + b^2 \quad \rightarrow (a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$a^2 - b^2 = (a - b) \cdot (a + b)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Zapisać podane wyrażenie w postaci sumy algebraicznej

$$(2x + 3y)^3 = (2x)^3 + \underline{3(2x)^2 \cdot 3y} + 3 \cdot (2x) \cdot (3y)^2 + (3y)^3 = 8x^3 + \cancel{36} 36x^2y + 54xy^2 + 27y^3$$

$$(3 - 2\sqrt{3})^3 = 3^3 - 3 \cdot 3^2 \cdot (2\sqrt{3}) + 3 \cdot 3 \cdot \underbrace{(2\sqrt{3})^2}_{12} - (2\sqrt{3})^3 =$$

$$= \underline{27} - 54\sqrt{3} + \underline{108} - 2^3 \cdot (\sqrt{3})^3 =$$

$$= 135 - 54\sqrt{3} - 24\sqrt{3} = 135 - 78\sqrt{3}$$

$$\left\{ \begin{aligned} (\sqrt{3})^3 &= \underbrace{\sqrt{3} \cdot \sqrt{3}}_3 \cdot \sqrt{3} = 3\sqrt{3} \end{aligned} \right.$$

$$\left(\frac{\sqrt{2}}{2} + 1\right)^3 = \dots = \frac{10 + 7\sqrt{2}}{4} \quad \frac{\text{licznik}}{\text{mianownik}}$$

$$\left(\frac{\sqrt{2}}{2}\right)^3 = \frac{(\sqrt{2})^3}{2^3} = \frac{\sqrt{2^3}}{8} = \frac{\sqrt{8}}{8}$$

$\parallel$
 $\frac{2\sqrt{2}}{8}$

Upność uproszczenia:

$$(x-3)(x^2+3x+9) = x^3 + \cancel{3x^2} + \cancel{9x} - \cancel{3x^2} - \cancel{9x} - 27 = x^3 - 3^3 = x^3 - 27$$

$$(2x+3)(4x^2-6x+9) = 8x^3 - \cancel{12x^2} + \cancel{18x} + \cancel{12x^2} - \cancel{18x} + 27 = 8x^3 + 27$$

$(2x)^3 + 3^3$

$$1) a^2 + b^2 \geq 2(a+b-1)$$

$$2) a^2 + b^2 \geq 4(a+b-2)$$

$$3) \frac{2}{a+b} \geq \frac{a+b}{a^2+b^2} \text{ dla } a, b > 0$$

Udowodnij, że dla dowolnych a i b

$$1) a^2 + b^2 \geq 2(a+b-1)$$

$$2) a^2 + b^2 \geq 4(a+b-2)$$

$$3) \frac{2}{a+b} \geq \frac{a+b}{a^2+b^2} \text{ dla } a, b > 0$$

$$\begin{aligned} 2x + 5 &= 3 \\ 2x &= -2 \\ x &= -1 \end{aligned}$$

Ad. 1.

Złożenie: $a, b \in \mathbb{R}$

$$\text{Teza: } \underline{a^2 + b^2 \geq 2(a-b-1)} \Leftrightarrow \underline{a^2 + b^2 - 2(a-b-1) \geq 0}$$

Trzeba zapisać wyrażenie $a^2 + b^2 - 2(a-b-1)$ w formie postaci,
żeby „było wiadome”, że jest ~~nieujemne~~ nieujemne.

Dowód:

$$\begin{aligned} L &= a^2 + b^2 - 2(a-b-1) = a^2 + b^2 - 2a - 2b + 2 = \\ &= \underbrace{a^2 - 2a + 1}_{\geq 0} + \underbrace{b^2 - 2b + 1}_{\geq 0} = (a-1)^2 + (b-1)^2 \geq 0 \quad \text{c.n.u.} \quad \text{QED} \end{aligned}$$

$$\text{Ad. 2} \quad a^2 + b^2 \geq 4(a+b-2)$$

Zał. $a, b \in \mathbb{R}$

$$\text{Teza: } a^2 + b^2 - 4(a+b-2) \geq 0$$

$$\begin{aligned} \text{Dowód: } L &= a^2 + b^2 - 4(a+b-2) = a^2 + b^2 - 4a - 4b + 8 = \\ &= \underbrace{a^2 - 4a + 4}_{\geq 0} + \underbrace{b^2 - 4b + 4}_{\geq 0} = (a-2)^2 + (b-2)^2 \geq 0 \end{aligned}$$

bo Produkt links fast weniger.
C.N.U.

Ad. 3

$$\frac{2}{a+b} \geq \frac{a+b}{a^2+b^2} \text{ d.h. } a, b > 0$$

Ziel. $a, b > 0$

Tezo. $\frac{2}{a+b} \geq \frac{a+b}{a^2+b^2} \Leftrightarrow \frac{2}{a+b} - \frac{a+b}{a^2+b^2} \geq 0$

Dowd: $\frac{2}{a+b} - \frac{a+b}{a^2+b^2} = \text{P.D.}$

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